Prior Work and Its Connection to Our HNC/MD Study

Entropy, diffusion, and structural diversity in charged mixtures

EntropicFluids collaboration

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Abstract

This review collects the prior work discussed in connection with the EntropicFluids study: excess-entropy scaling in simple and charged mixtures, high-entropy electrolytes, and Voronoi/graph estimates of structural entropy. It distinguishes what these studies establish from what remains to be tested in our model mixtures. The general idea that multicomponent electrolytes can show enhanced ion mobility already has direct precedent. The potential value of the present study lies in its controlled comparisons, independent thermodynamic-integration benchmark, and distinction between species labels, interaction diversity, and probe-ion motion.

1 Scope and terminology

This is an annotated literature assessment rather than a systematic review or a claim of priority. References include the studies discussed previously and supporting methodological sources. The companion `voronoi_entropy_work_plan.tex` gives the proposed analysis protocol; this document supplies the scientific context.

Three entropy notions must remain distinct. Ideal compositional mixing has entropy per particle $s_{\mathrm{mix}}^{\mathrm{id}} = -k_B \sum_i x_i \ln x_i$. Thermodynamic excess entropy is

$$s_{\mathrm{ex}} = \frac{S - S_{\mathrm{id,mix}}(T, \rho, \{x_i\})}{N}, \tag{1}$$

so that the ideal mixing term cancels against the same-composition reference. Finally, a graph or local-environment entropy describes the diversity of a chosen structural representation. A solvation-state entropy is likewise conditional on the chosen definition of a solvation state. These quantities can correlate without being interchangeable. In the glass literature, configurational entropy often refers specifically to basin multiplicity after separating vibrational entropy, which is another distinct target.

2 Excess entropy and diffusion in mixtures

2.1 Rosenfeld: foundations and charged-mixture precedent

Rosenfeld's original work [1] motivates an approximate relation between reduced transport and excess entropy. For self-diffusion, write

$$D_i^{\mathrm{R}} = D_i \rho^{1/3} \sqrt{m_i/(k_B T)}, \qquad \ln D_i^{\mathrm{R}} \simeq a_i + b_i s_{\mathrm{ex}}/k_B. \tag{2}$$

This representation is useful empirically; it does not imply that changing any contribution called entropy must accelerate every species.

Rosenfeld, Nardi, and Zinamon [2] developed a corresponding-states hard-sphere description of diffusion in binary dense-plasma mixtures, tested against molecular dynamics of asymmetric ionic mixtures. Charged-mixture entropy–diffusion connections therefore have a long history. Their

plasma setting differs from our explicit neutral solvent with oppositely charged probe ions, but a broad claim of introducing entropy scaling to charged mixtures would be inappropriate.

Dyre's perspective [14] explains the connection with hidden scale invariance and isomorph theory, while reviewing failures of universality. Its practical implication here is that a successful fit over a restricted composition range is evidence of an empirical relation, not an exact law covering changes in Hamiltonian or composition.

2.2 Individual-species diffusion: Krekelberg et al. (2009)

Krekelberg et al. [3] studied tracer diffusivities in hard-sphere and Gaussian-core model fluids, including mixtures with size asymmetry and nonadditive interactions. Their generalized Rosenfeld reduction uses interaction information and dilute-limit thermodynamic and transport behavior. It captures many composition effects, but encounters limitations when the component diffusivities decouple strongly.

This is the main precedent for our distinction between the composition-weighted mean self-diffusion and divalent-ion self-diffusion. Ordinary Rosenfeld units and the generalized reduction are different constructions. Our measurements should not be described as a test of their generalized prediction unless its required dilute-limit quantities are actually calculated. A weak species-level correlation with one bulk entropy is not intrinsically inconsistent with mixture scaling research.

2.3 A predictive mixture framework: Schmitt et al. (2025)

Schmitt, Hasse, and Stephan [4] combine entropy scaling, limiting pure-component and infinite-dilution transport properties, and mixing rules to describe self-diffusion and mutual diffusion in binary mixtures. Mixture entropy is supplied by an equation of state. They test model and real-fluid systems, and emphasize that a general single-variable mixture dependence does not persist.

Their work shows how entropy can remain useful even when it is insufficient alone. It also reinforces a necessary distinction: our $D_{\mathrm{net}} = \sum_i x_i D_i$ is a weighted mean of self-diffusion coefficients, not a mutual-diffusion coefficient or an electrical conductivity. A direct comparison with their predictive framework would require additional mixture transport information and an extension beyond the binary demonstrations.

3 High-entropy electrolytes: the closest application precedents

3.1 Wang et al.: multisalts and enhanced lithium mobility (2023)

Wang et al. [5] compare a four-salt lithium electrolyte with single-salt counterparts at matched total salt concentration. Experiments and MD support altered solvation and enhanced lithium mobility, alongside improved battery performance. The paper explicitly recognizes the challenge of separating the effect of entropy from the properties of the selected chemical components.

This closely matches the motivation of our simulations. However, chemical substitution changes interactions as well as compositional diversity. The reported association is therefore not, by itself, evidence that ideal mixing entropy independently causes a mobility increase. Our identical-interaction label control and controlled distribution of solvent interactions address that attribution problem in a simpler setting.

A related paper by Wang et al. [15] examines entropy-driven multisalt electrolytes, including solubility, solvation, interphase stability, and lower-temperature operation. It broadens the application context but is a separate paper from Ref. [5]; their findings should not be combined into a single bibliographic attribution.

3.2 He et al.: assessment from the supplied full article (2026)

The supplied ten-page article [6] permits a more specific assessment than the earlier abstract-level discussion. The principal evidence is in Sec. 2.1 (journal pp. 683–685), Sec. 2.2 (pp. 685–686), and Fig. 3. The supplementary information was not supplied or successfully retrieved; claims below about reporting and uncertainty refer to the main article, not to an assertion that the supplementary methods contain no additional checks.

3.2.1 Design and strengths

The authors compare four formulations, EL1, EL2, EL3, and EL5, with one, two, three, and five anion species, respectively. Total lithium-salt concentration remains 1.0 M, with the same carbonate solvent base and 5% FEC additive. EL1 contains 1.0 M $LiPF_6$; EL2 has 0.6 M $LiPF_6$ and 0.4 M LiTFSI; EL3 has 0.6 M $LiPF_6$, 0.2 M LiTFSI, and 0.2 M LiDFOB; EL5 has 0.6 M $LiPF_6$ plus 0.1 M each of LiTFSI, LiDFOB, LiFSI, and $LiNO_3$. Matching total salt concentration is a useful control. With monovalent ions it also matches nominal ionic strength under complete dissociation; actual association need not match.

The combination of Raman spectra, MD coordination statistics, representative cluster calculations, and electrochemistry supports an altered solvation environment. The solvent contribution to Li coordination decreases and several anions participate directly. The application results on nitrate solubility and electrode interphases add practical value. These strengths should be assessed separately from the stronger mechanistic claim that entropy determines diffusion.

Changing salt identities and fractions also changes binding, geometry, association, and solvent exchange. Thus the series is a meaningful chemical design study, but it does not isolate component number or an entropy change as an independent intervention. Fixed molarity alone does not fix particle number density, force-field interactions, viscosity, or molecular packing.

3.2.2 The two entropy measures have different targets

Their Eq. (2) defines a solvation-state Shannon entropy,

$$S_{\mathrm{sc}} = -R \sum_a p_a \ln p_a, \tag{3}$$

which rises from 32.5 to 40.4 J mol^{-1} K^{-1} between EL1 and EL5. This usefully summarizes the diversity of the selected Li-centered solvation classes. Its value depends on the class definitions, neighbor cutoffs, chemical labels, and sampling of rare classes. Adding anion identities expands the set of possible chemical classes. A composition-preserving random-label reference, or a comparison using merged chemical classes, would help separate this combinatorial diversity from changed local ordering. That suggestion does not imply that the real anions are mechanically identical.

S_{sc} is not the total liquid entropy: it omits within-class information and correlations between overlapping Li environments. Its natural reference is a mole of sampled Li environments, whereas our TI entropy is normalized per particle of the entire mixture. Their Eq. (1), the ideal mixing formula, is likewise not a general expression for full configurational entropy in an interacting molecular liquid. None of these quantities should be added together without a consistent decomposition and normalization.

For the second measure, the prose on p. 686 specifies Li–anion and Li–solvent pair correlations in the local solvation environment. Their Eq. (4), however, has the form of the standard all-species pair-entropy sum:

$$\frac{s_2}{k_B} = -2\pi\rho \sum_{\alpha\beta} x_\alpha x_\beta \int_0^\infty [g_{\alpha\beta}(r) \ln g_{\alpha\beta}(r) - g_{\alpha\beta}(r) + 1]\, r^2\, dr. \tag{4}$$

Visual inspection confirms that the printed bracket is $g \ln g - [g - 1]$: its sign is correct. The issue is the estimator's scope, not a sign error in this equation.

If only the specified Li-containing pairs are retained, the result is a selected contribution to s_2, rather than the total mixture s_2 or s_{ex}. If the implemented calculation actually includes all species pairs, the prose needs clarification. The main article does not resolve this ambiguity sufficiently for reproduction. The atom-versus-molecule species convention, total density, mole fractions, pair multiplicities, radial cutoff, and per-Li versus per-total-particle normalization must also be established before comparing values across changing anion chemistries. For molecular models these conventions matter even at the same salt molarity.

The assertion that the pair term accounts for more than 90% of excess entropy is not independently benchmarked against a total-entropy calculation for these electrolytes in the main article. Such a fraction is not a general identity. It is particularly insufficient to equate a selected local pair sum with full excess entropy. A TI or other independently validated thermodynamic reference is needed to measure the missing contribution.

3.2.3 What the reported transport correlation establishes

The main text reports an increase in Li self-diffusion from 1.42×10^{-7} to 2.38×10^{-7} cm^2 s^{-1}, approximately 68%, while the stated entropy proxy changes from approximately -0.071 to -0.068 in units of $S_2/(Nk_B)$. Fig. 3d plots $\log_{10} D_{\mathrm{Li}}$ against this pair proxy for four formulations. It is a semilog correlation, not a straight-line test of untransformed D. Four means and a fitted slope/intercept provide only two residual degrees of freedom; independent replica uncertainty, fit confidence intervals, and held-out predictions are not displayed in that figure. This observation does not establish whether such checks were performed elsewhere.

Their Eq. (5) uses dimensional D, whereas conventional Rosenfeld scaling uses $D^{\mathrm{R}} = D\rho^{1/3}\sqrt{m/(k_B T)}$ and an intensive excess entropy. At identical temperature, Li mass, and particle number density, ratios of D and D^{R} would coincide. Fixed salt molarity does not alone establish the last condition for these changing molecular compositions. A common, documented density convention and explicit normalization of the entropy coordinate are therefore needed. This limitation does not mean that the plotted trend is spurious; it limits its interpretation as a classical Rosenfeld validation or a comparison with our bulk-entropy slopes.

The data support a correlation between formulation, selected structural statistics, and Li mobility. They do not independently determine whether entropy, altered binding, viscosity, association, or exchange kinetics is the causal factor. Improvements in interfacial resistance and battery cycling support the electrolyte design, but also depend on changed interphase chemistry and are not independent validations of a bulk self-diffusion law.

3.2.4 Relation to our HNC/MD work and useful follow-up

The closest comparison is with an ion-centered pair or neighborhood *descriptor*, rather than with our total TI entropy. Their selected local coordinate may be useful even when bulk entropy predicts probe mobility imperfectly. It also motivates the proposed charge/species-aware Voronoi analysis, with a chemical-composition reference.

Our independent TI benchmark and exact label null address attribution and normalization questions that the main article leaves open. Our controlled interaction-amplitude comparison is also more restrictive, although it still changes local energetics and friction and does not prove entropy causation. Conversely, their realistic molecular chemistry and experimental validation provide application evidence that our coarse-grained study does not offer. Neither study establishes a universal species-level entropy–diffusion law.

The most informative checks would be:

1. Reproduce the published local pair statistic with an explicit species, weighting, density, and radial-range convention, then compare it with a full all-pair s_2 and an independent total-entropy benchmark. For our systems, calculate the corresponding divalent-ion-centered contributions and compare both their increments and the remainder $s_{ex} - s_2$ against TI.

2. Repeat the semilog fits with reduced diffusion, independent replicas, MSD-window checks, uncertainty in both coordinates, and additional compositions or state points reserved for prediction. A four-formulation fit is a useful observation, not a strong universality test.

3. Test solvation-class convergence and composition-preserving permutation references. Compare topology, charge-only classes, and full chemical classes before interpreting a larger state alphabet as increased thermodynamic entropy.

4. Evaluate association and residence times, viscosity or friction proxies, and species-resolved motion. Controlled substitutions at matched nominal ionic strength and concentration can help identify the mechanism, while retaining a clear account of the interactions that actually change.

Assessment. This is a relevant and experimentally supported electrolyte-design study with a useful local structure–mobility correlation. The main article's evidence is insufficient for its stronger claim of strict Rosenfeld scaling with total thermodynamic excess entropy or proof of entropy-driven diffusion. It is a close precedent for our motivation and a valuable source of local descriptors, rather than a quantitative validation of the same entropy observable we obtain by HNC and MD/TI.

4 Voronoi and graph-based structural entropy

4.1 Local topology and its implementation

Lazar, Han, and Srolovitz [8] describe a Voronoi-topological framework for classifying local atomic structure. The VoroTop implementation [9] makes this approach accessible for atomistic configurations. Graph connectivity resolves distinctions that coordination numbers or simple face-count indices can miss. These sources support the structural representation; they do not by themselves validate an absolute thermodynamic-entropy estimator for our mixtures.

4.2 Yoon et al.: topological entropy and transport (2019)

Yoon et al. [7] classify complete Voronoi-cell graph topologies using Weinberg vectors and calculate a Shannon entropy from their frequencies. For repulsive soft-sphere fluids, they compare this descriptor with TI excess entropy and reduced transport. They use an ideal-gas topological reference and address unseen structural types. The relation to thermodynamic entropy is interaction dependent, and neighboring-cell correlations remain important.

This is the closest methodological precedent for testing topology against our TI and diffusion results. It motivates a structural benchmark rather than assuming that the raw local Shannon entropy equals s_{ex}/k_B. In our mixtures, a further question is whether geometry alone misses changes in the charge and species identities of neighboring particles. That is an inference motivating our proposed colored-neighborhood analysis, not a result already established by this reference.

4.3 Zhou and Milner: cluster-entropy extrapolation (2016)

Zhou and Milner [10] classify Voronoi-derived clusters using graph isomorphism. They estimate subsystem structural entropy and its growth with particle number; the slope gives an entropy per particle. Their hard-particle results are consistent with previous configurational-entropy estimates from thermodynamic integration.

This goes beyond a histogram of isolated cell types by analyzing larger structures. Its transfer to attractive charged liquids requires separate validation: graph-equivalent configurations can have different distances, energies, and within-class phase-space volumes. Sampling rare graph classes also becomes harder as cluster size grows. The companion plan therefore makes extrapolation conditional on adequate class coverage, a stable entropy-growth regime, and an explicit definition

of the entropy target. A stable graph-entropy slope alone is not proof of equality to liquid excess entropy.

4.4 Molecular-liquid applications

Abdulkareem U and Madhurima [11] use Voronoi entropy to characterize hydrogen-bonded liquids and binary alcohol mixtures. This establishes broader use of structural-diversity measures in molecular mixtures. It is not a direct demonstration that they provide absolute excess entropy or a universal ion-diffusion predictor. The issue year is 2024; the article became available online in November 2023.

5 The 2PT comparison and its limits

Sun et al. [12] reanalyze the two-phase thermodynamic approach to liquid entropy. Lai et al. [13] extend 2PT treatment to mixtures. These are the methodological sources for our corrected classical spectral assessment, not independent validations of the present charged mixtures.

Our rehabilitation tests pass internal normalization and reference checks, but retain composition-dependent discrepancies against TI. In particular, the primary corrected estimate overstates the controlled entropy increment with monovalent background by approximately $0.036k_B$ per particle. Consequently TI remains our primary entropy coordinate. A static Voronoi/graph comparison has the advantage that it does not take a diffusion coefficient as input, whereas the 2PT gas/solid partition is informed by diffusion. Agreement between 2PT entropy and diffusion therefore has a different interpretation from an independently calculated static-structure correlation.

6 How the prior work connects to our HNC/MD study

6.1 The role of HNC and the independent MD test

Our workflow uses hypernetted-chain (HNC) calculations to screen charge, composition, and solvent-interaction candidates for higher excess entropy, then tests selected states with independent MD thermodynamics and transport. The successful entropy ordering is a validation of the screening for these candidates, not an exact prediction of their dynamics. In the completed study, HNC absolute entropies are more negative than TI values by approximately 0.064–$0.093k_B$ per particle; controlled diversification increments agree more closely: about 0.1107–$0.1108k_B$ from HNC versus 0.1032–$0.1033k_B$ from MD/TI.

This makes the thermodynamic and kinetic steps separately assessable. Agreement of HNC with TI is distinct from agreement of an entropy-based mobility forecast with measured D_i. HNC does not directly provide self-diffusion coefficients in the present implementation. A species-specific mobility prediction requires an additional dynamical assumption or model. The screened candidate set also does not establish a global composition optimum.

6.2 Connection to the five closest entropy–diffusion precedents

The links below are our assessment of how the cited studies inform this project; they are not claims that those authors examined our particular controls or divalent-solvent model.

Prior work	Connection and distinction for the present calculations
Rosenfeld, Nardi, and Zinamon (1995) [2]	Establishes an early charged-mixture precedent for entropy-based diffusion modeling. Our neutral solvent, oppositely charged probes, and physical solvent-diversification controls address a different setting; charged-mixture scaling itself is not new.

Prior work	Connection and distinction for the present calculations
Krekelberg et al. (2009) [3]	Provides the main context for tracer mobility and component decoupling. It motivates comparing bulk and probe responses separately. Our ordinary reduced-diffusion plots do not implement their generalized dilute-limit scaling, so disagreement with a simple bulk-entropy correlation would not by itself falsify their model.
Wang et al. (2023) [5]	Directly motivates the multispecies-electrolyte question: can composition and solvation diversity improve ion motion? Our simpler model adds a label null and matched mean bare interaction amplitude. These help examine attribution, while actual energy, friction, and local solvation still change.
Schmitt, Hasse, and Stephan (2025) [4]	Shows how mixture entropy can support diffusion prediction when supplemented by limiting transport properties and mixing rules. This supports using HNC as thermodynamic input while recognizing that it supplies only part of a predictive transport model. Their binary framework is not directly applied here to our multicomponent charged mixtures.
He et al. (2026) [6]	The closest broad application precedent, now assessed from the supplied full article (Sec. 3.2). Its four-formulation fit uses a Li-centered pair-entropy proxy; our HNC/MD comparison uses total TI entropy. Its chemistry and experiments add application evidence, while our independent entropy benchmark and label control test different methodological questions.

6.3 Would using pair entropy change our conclusions?

The existing fine-grid HNC output already contains the *full all-species* pair entropy. It therefore permits a preliminary comparison without new MD or a fitted entropy model. Table 2 uses these existing values. This is not an RDF-based MD pair-entropy benchmark: the latter still needs to be computed independently from the trajectories. In particular, subtracting HNC s_2 from MD/TI s_{ex} would mix two methods and would not measure an MD higher-order entropy contribution.

Table 2: Existing HNC pair and full-entropy increments, in k_B per particle. Differences use the four-decimal fine-grid summary; percentages are descriptive, not uncertainty estimates.

Contrast	$\Delta s_2^{\mathrm{HNC}}$	$\Delta s_{\mathrm{ex}}^{\mathrm{HNC}}$	Pair fraction
L5–U4	0.0000	0.0000	—
M5–U4	0.0226	0.0332	68%
H5–U4	0.0750	0.1107	68%
H7–U6	0.0751	0.1108	68%
B5–B3	−0.0025	−0.0042	60%

For all seven mechanically distinct states, HNC s_2 and HNC s_{ex} have the same ordering. The identical-label U4/L5 comparison remains an exact null. Physical solvent diversification still increases entropy; monovalent addition B3→B5 still decreases it slightly. Thus *the existing HNC pair data support the same qualitative conclusions*, including the broad association with increased diffusion of both the mean liquid and the divalent probes. The diffusion values and their uncertainties do not change when the entropy coordinate is replaced, so the small controlled probe effect remains provisional.

The quantitative interpretation does change. HNC s_2 accounts for only approximately 75.6–76.2% of the magnitude of HNC s_{ex} across these states. More relevantly, it captures only about 68% of the controlled solvent-diversification increments. The HNC remainder $s_{\mathrm{ex}}^{\mathrm{HNC}} - s_2^{\mathrm{HNC}}$ increases

by about $0.0357k_B$ for both strong splits. These are differences within the HNC approximation, not independently measured exact many-particle entropy terms. Consequently, pair entropy alone would understate the HNC entropy increment by about one third and change fitted entropy–diffusion slopes. It would not establish quantitative equality to TI or a universal Rosenfeld relation. The frequently invoked greater-than-90% approximation is not supported by these particular HNC results; this observation does not determine its accuracy in the different molecular electrolytes of He et al.

Finally, this all-species s_2 comparison does not validate the selected Li-centered statistic used in the main-text description of He et al. Selected probe-centered contributions can behave differently from the full pair sum. A useful next check is to calculate both from the same MD RDFs, with identical density, normalization, and integration conventions, and compare their controlled increments against TI. Numeric sources and the preliminary comparisons are recorded in `doc/hnc_pair_entropy_comparison.json`.

6.4 What this implies for the original question

An increase in the number of components can increase ideal mixing entropy without changing forces, as our U4/L5 control demonstrates. To affect mobility, physical diversification must change the mechanical environment; that change may simultaneously alter excess entropy and diffusion. Our broad mixture trends are consistent with an entropy–mobility association, including for the divalent probes. They do not show that an ideal mixing contribution causes faster motion, or that one bulk entropy determines each species' diffusivity. The smaller controlled probe response remains less certain than the corresponding entropy and solvent-dominated mean-diffusion changes.

7 Assessment of the present study's contribution

The following statements concern our completed calculations, not the external papers. All states have $N = 4000$, $T^* = 2$, $\rho^* = 0.7$, and 3% of each divalent sign, with three replicas. The TI reference is the ideal mixture of the same composition. The LJ Hamiltonian is held fixed along each density integration path.

Finding or control	Interpretation in light of prior work
Identical-label splitting	U4 and L5 have identical interactions. Their formal ideal mixing entropy differs, while diffusion is identical and TI entropy agrees within sampling uncertainty. Component labels alone supply no mechanical acceleration.
Broad design changes	Higher-excess-entropy solvent designs increase pooled divalent self-diffusion by about 18–20% relative to the baselines. This supports an association but also changes mean interactions and local friction.
Controlled interaction diversity	At fixed charged composition, nominal sizes, masses, ionic strength, and mean bare LJ amplitude, strong solvent splitting raises TI entropy by about $0.103k_B$. Probe diffusion increases by about 5% in point estimates, with unresolved replica or fitting-window uncertainty. Actual energy and friction are not held fixed.
Species versus bulk response	The controlled increase in weighted mean self-diffusion is about 8–9%, larger than the probe response. Broad entropy–diffusion trends exist for both, but a single bulk entropy does not determine each species' mobility.

Finding or control	Interpretation in light of prior work
Independent entropy benchmark	TI permits a direct assessment of HNC, 2PT, and prospective graph descriptors. Passing internal consistency tests is distinct from agreeing quantitatively with TI across compositions.

These controls provide a plausible contribution alongside the existing literature. The appropriate claim is a controlled examination of the entropy–mobility association and its limits in model charged mixtures. The present evidence does not demonstrate causation by entropy, universality across mixtures, or a converged small enhancement of probe mobility under every controlled intervention.

For the Voronoi follow-up, the main opportunity is to distinguish local packing from chemical organization around the probes, and to determine whether either adds explanatory value beyond TI and pair entropy. Even a negative benchmark can be useful if a descriptor tracks solvent-dominated mean diffusion but misses probe behavior. A focused pilot is justified; cluster extrapolation should follow only after its sampling requirements and entropy interpretation are demonstrated.

Local sources and reproducibility

The completed numerical evidence is in the study directory `ComponentEntropyStudy_20261030/`. Relevant files within it are:

- `results/conclusions.md`;
- `results/summary.json` and `results/diagnostics.json`;
- `two_pt_rehabilitation/report.md`.

The proposed analysis is documented separately in `doc/voronoi_entropy_work_plan.tex`. The prior-work assessment does not replace a full reading of the primary papers or establish that no equivalent controlled study exists elsewhere.

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